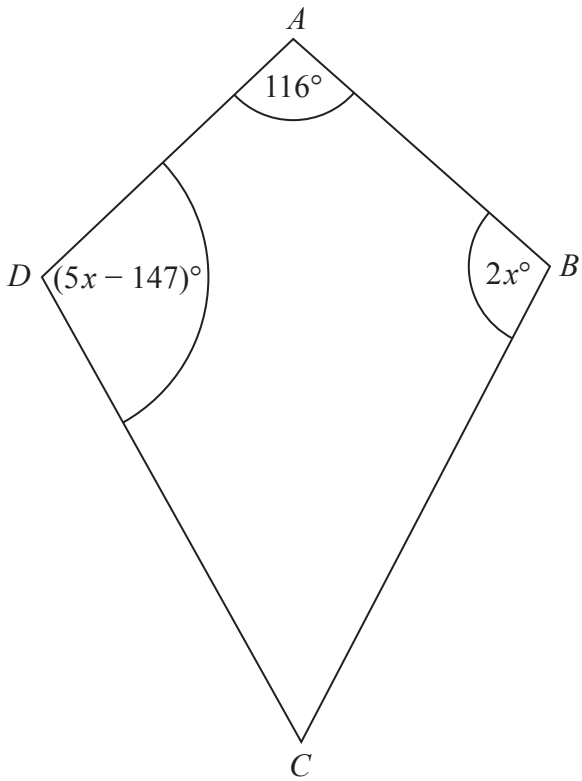
15 $ABCD$ is a kite with $AD = AB$



Find the size of the smallest angle of the kite.

Angles B and D are equal in a kite.

$5x - 147 = 2x$

$-2x \qquad -2x$

$3x - 147 = 0$

$+147 \qquad +147$

$3x = 147$

$\div 3 \qquad \div 3$

$x = 49$

1 mark
Equal opposite angles

1 mark
Solve: x = 49

$B = D = 2 \times 49 = 98°.$

$C = 360° - 116° - 98° - 98° = 48°.$

The smallest angle is $C = 48°$.

1 mark
Angles sum to 360°

1 mark
48°

°

(Total for Question 15 is 4 marks)